

Date	30 Jun 2026
Team ID	
Project Name	Food Ordering Behaviour & Customer Trends: A Structured Analysis of Choices and Habits
Phase	Project Development Phase

Project Development Phase — Document 1

Data Preprocessing Steps and Business Questions with Visualization

1. Data Preprocessing Steps

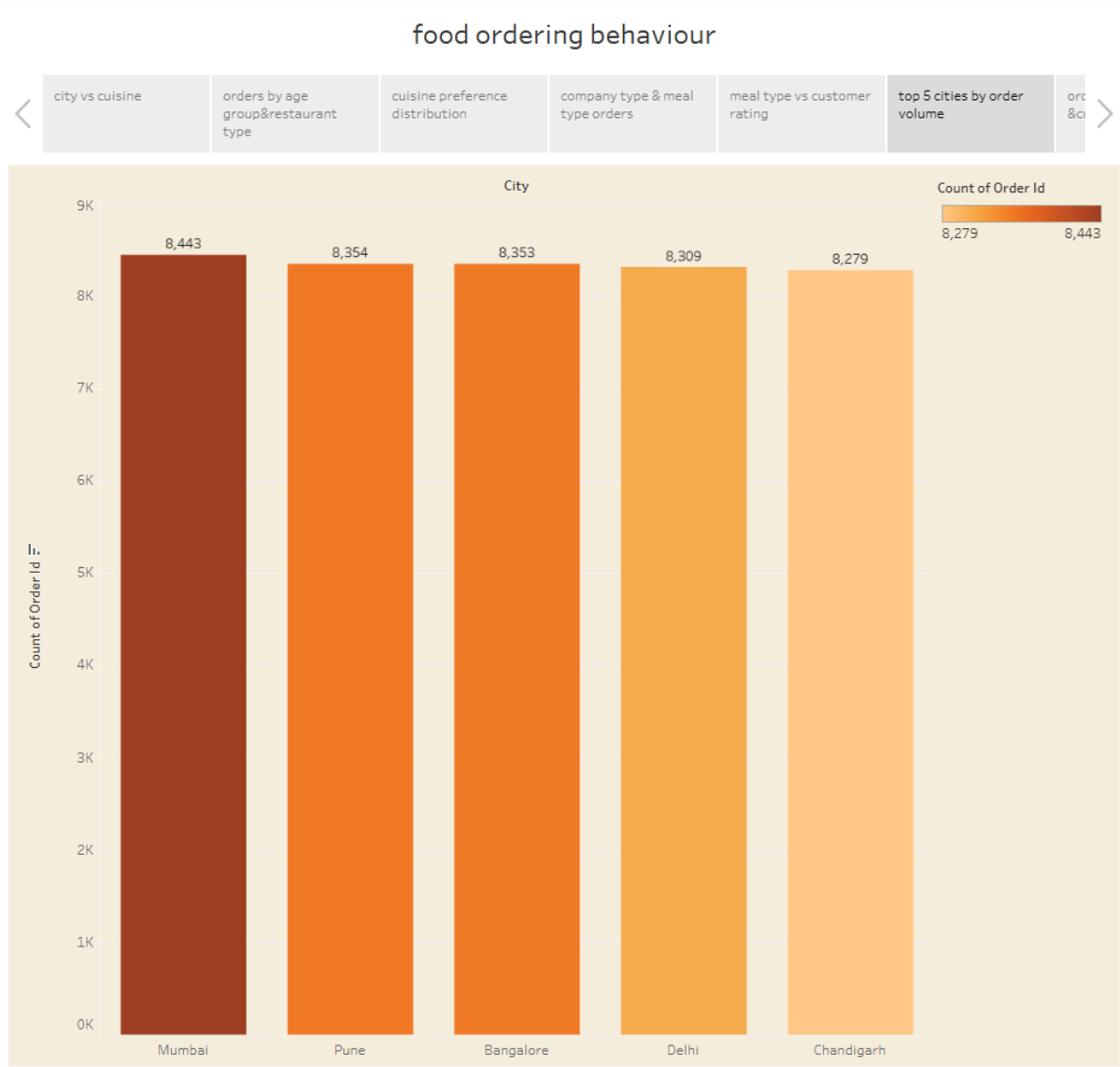
The raw dataset (food_ordering_behavior_dataset.csv) was inspected and prepared before visualization in Tableau:

- Schema validation: confirmed 50,000 rows across 19 columns with correct data types (integers for order_value, delivery_fee, rating_given, age, time_taken_to_order; text/categorical for city, cuisine, meal_type, etc.).
- Completeness check: 0 missing values and 0 duplicate rows found across all 50,000 records.
- Derived field — Age Group: bucketed 'age' into four bands (18-24, 25-31, 32-38, 39-44) for segment-level analysis.
- Derived field — Revenue: aggregated using SUM(order_value) for city/mealtype/cuisinelevel rollups.
- Derived field — Repeat Order Rate: computed as the share of orders where is_repeat_order = 'Yes'.
- Categorical standardisation: confirmed consistent category labels across city, cuisine, meal_type, restaurant_type, mood, hunger_level, company and rainy_weather fields prior to building calculated fields.
- Extract creation: the cleaned dataset was loaded into a Tableau Hyper Extract (.hyper) for fast query performance across all worksheets.

2. Business Questions & Visualizations

Q1. Which cities generate the highest order volume?

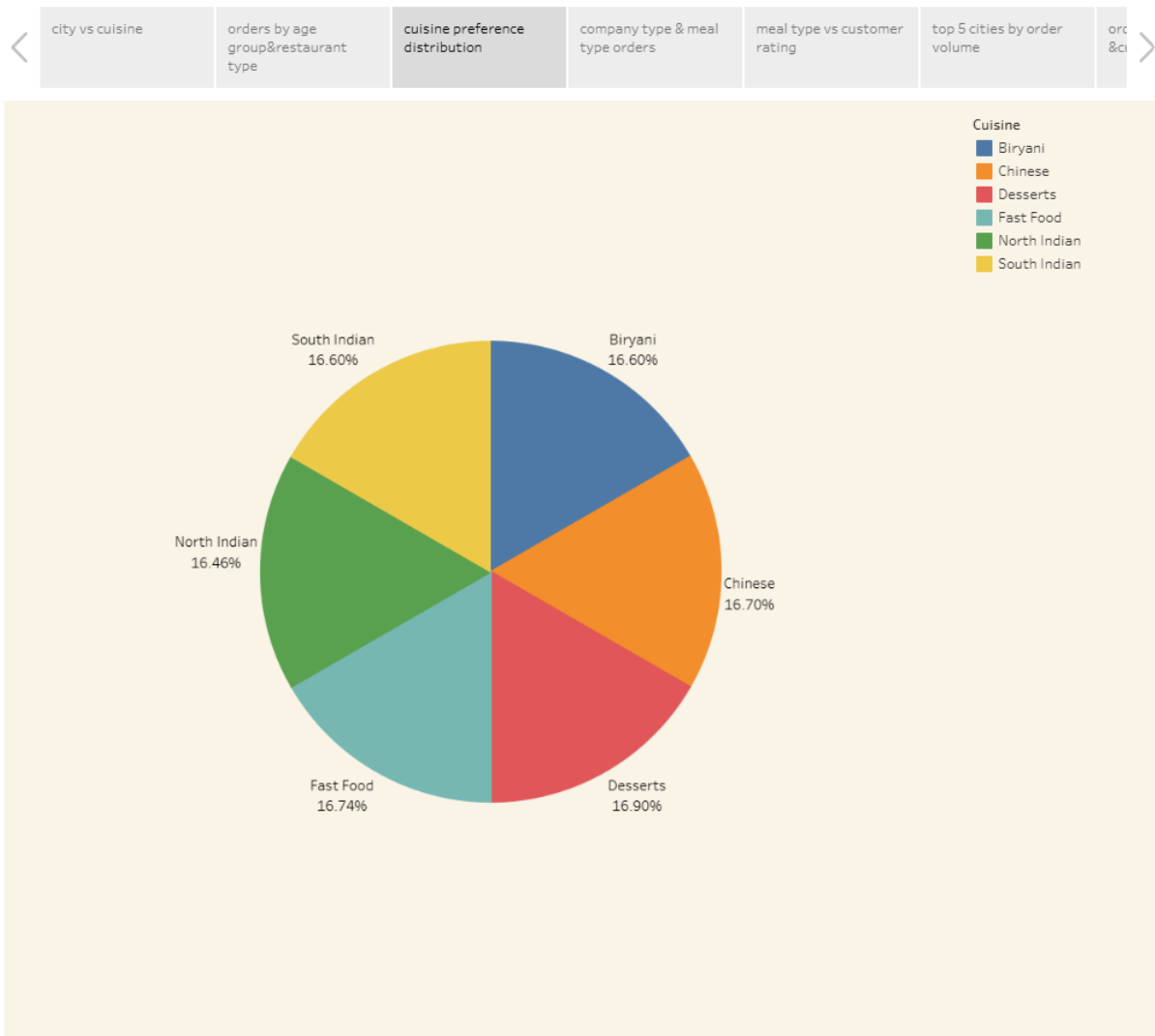
Mumbai leads with 8,443 orders, closely followed by the other five cities (Pune, Bangalore, Delhi, Chandigarh, Hyderabad), each contributing a broadly similar share — indicating demand is fairly evenly spread rather than concentrated in one region.



Q2. What cuisines do customers prefer?

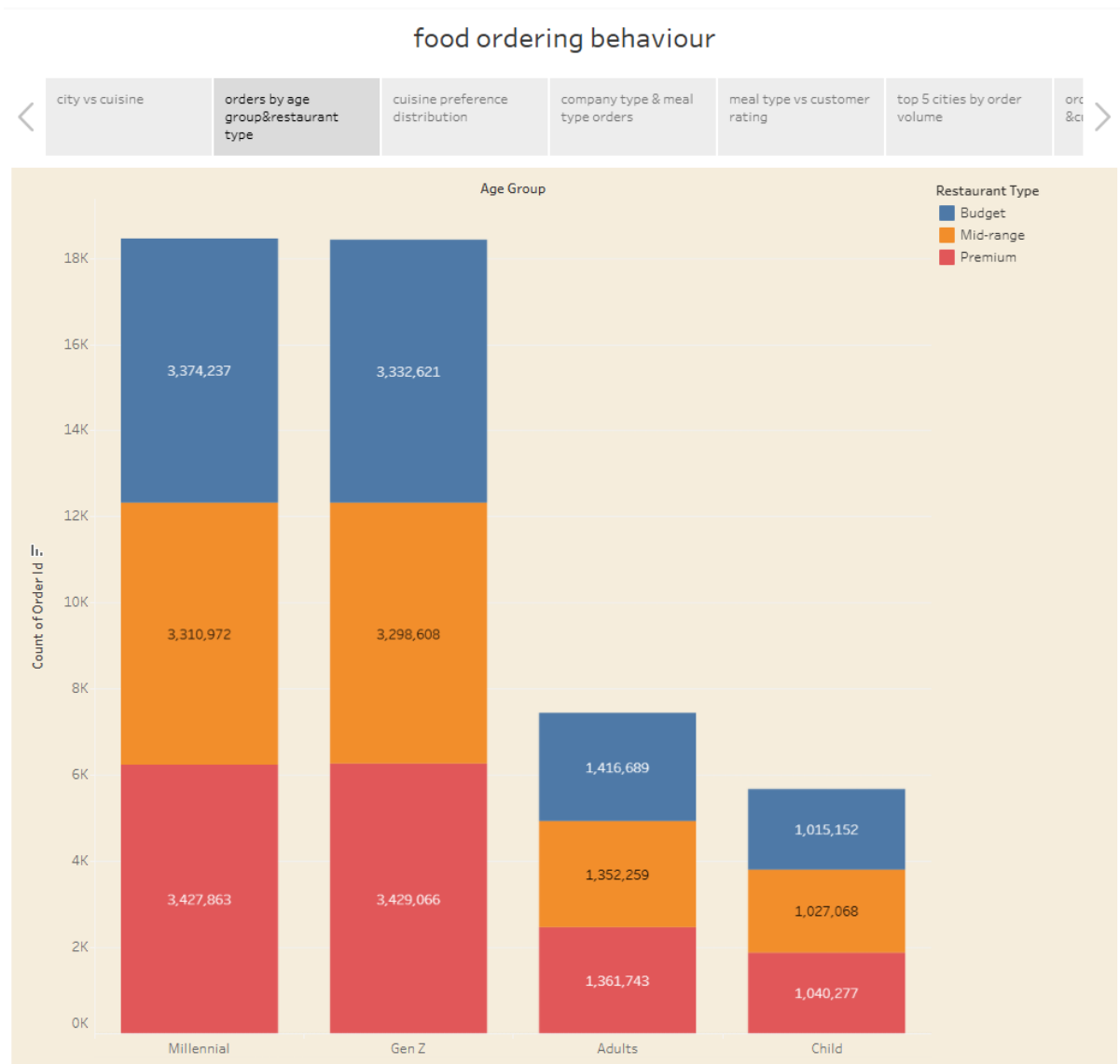
Preferences are close to evenly split across the six cuisines (Desserts ~17%, Fast Food/Chinese/South Indian/Biryani ~17% each, North Indian ~16%), suggesting no single cuisine dominates overall demand.

food ordering behaviour



Q3. How does order distribution vary by age group and restaurant segment?

All four age groups (18-24 through 39-44) order fairly evenly across Budget, Mid-range and Premium restaurant segments, with only slightly higher Premium share in the 39-44 group.



Q4. How does company type relate to meal type chosen?

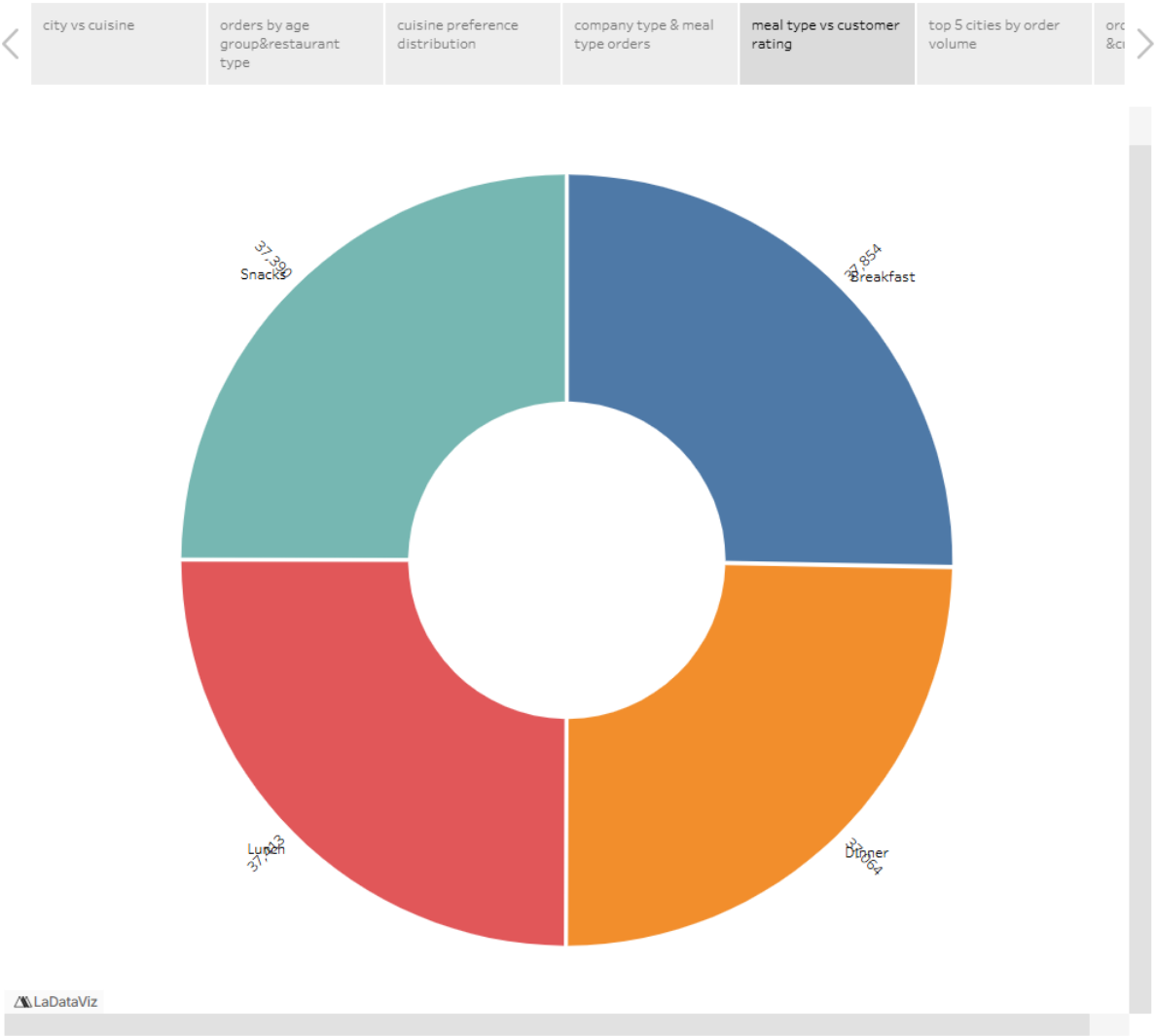
Order counts across company types (Alone, Family, Friends, Partner) are distributed similarly across all four meal types, without one company type dominating a specific meal occasion.



Q5. How do customer ratings vary by meal type?

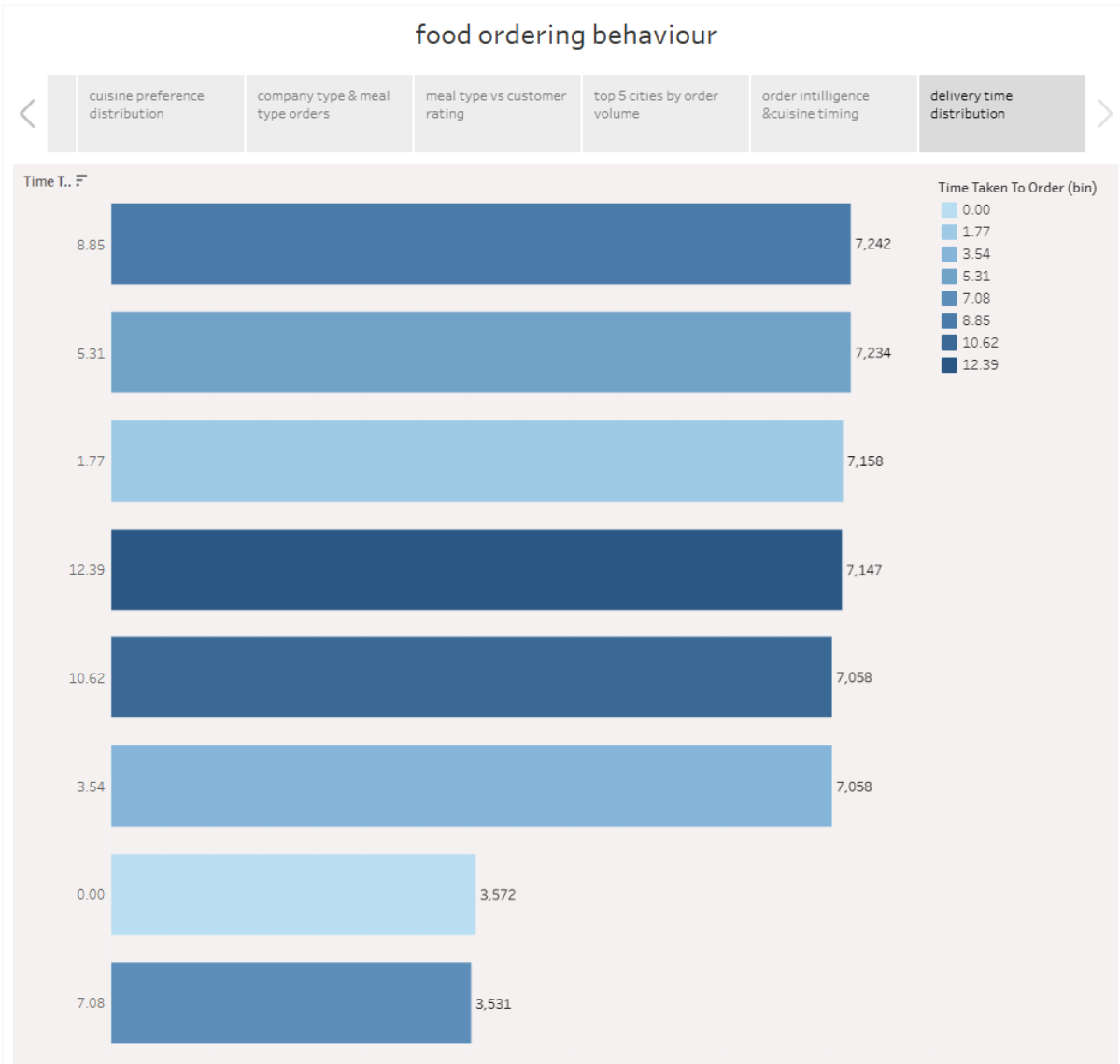
Average ratings are consistent across all meal types, clustering around 3.0 out of 5 — indicating a stable, moderate satisfaction level regardless of meal occasion.

food ordering behaviour



Q6. What does the order-time distribution look like?

Time taken to place an order is fairly uniform across the observed range (roughly 2–14 minutes), with no sharp spikes — suggesting a consistent ordering experience.



Q7. Which meal type contributes the most revenue?

Breakfast contributes marginally more total revenue (~₹69.2L) than Snacks, Lunch and Dinner, though all four meal types are closely matched.

